# Use Transit Gateway

## (LAB-M09-01)

**Overview**

In this lab, you are introduced to AWS Transit Gateway. Configure network routing between three Amazon VPC's located in two separate AWS Regions.
In this lab, you will connect the three different VPC(s) to host the web and db application.

**Objectives**

After completing this lab, you will be able to:

- Set up AWS Transit gateway.
- Configure Routing table.
- Enable communication between multiple VPC(s).

## Task 1: Create VPC and Web Server

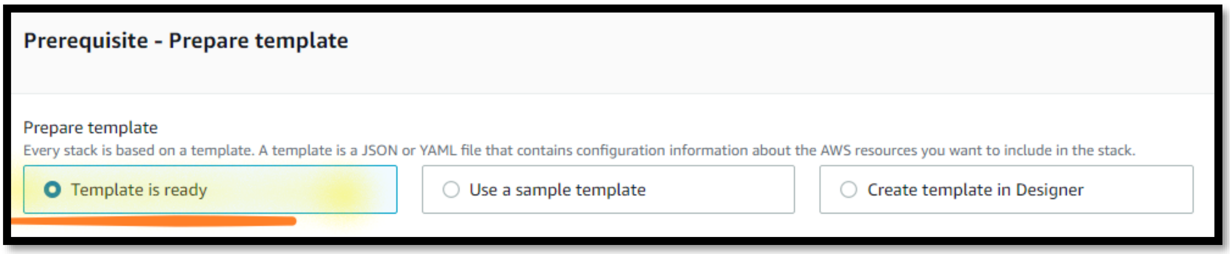
### Step 1: Deploy a VPC using CloudFormation

AWS CloudFormation model and provision AWS resources in your cloud environment in automated way.

**Note**

- In this task, CloudFormation provision Virtual Network, where you will deploy the resources.

1. In the **AWS Management Console**, on the **Services** menu, click **CloudFormation**.
2. Dropdown and select the **US East (N. Virginia)** region.
3. Click **Create stack**:
   a. In the **Prerequisite - Prepare template** section:
      i. **Prepare template**: Select **Template is ready**.

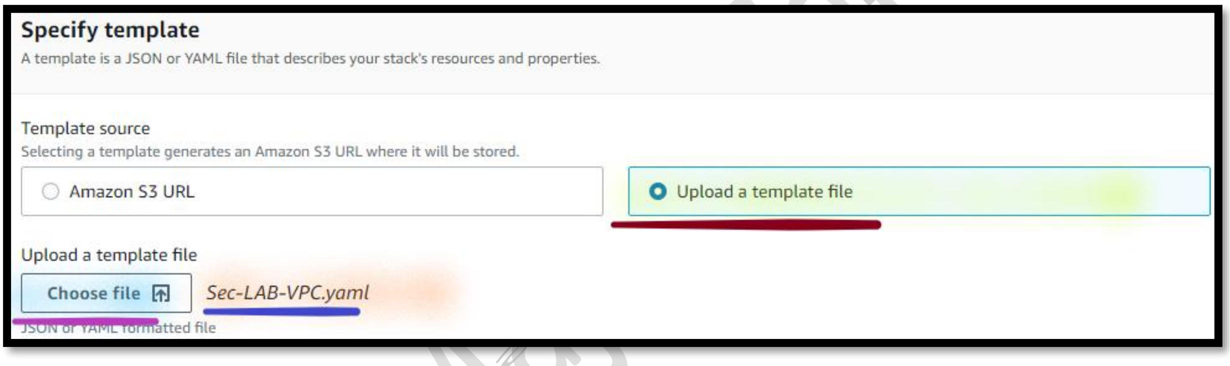



b. In the **Specify template** section:

i. **Template source**: Select **Upload a template file**.

ii. **Upload a template file**: Click on **choose file** then select the **LAB-VPC-01**.yaml file.

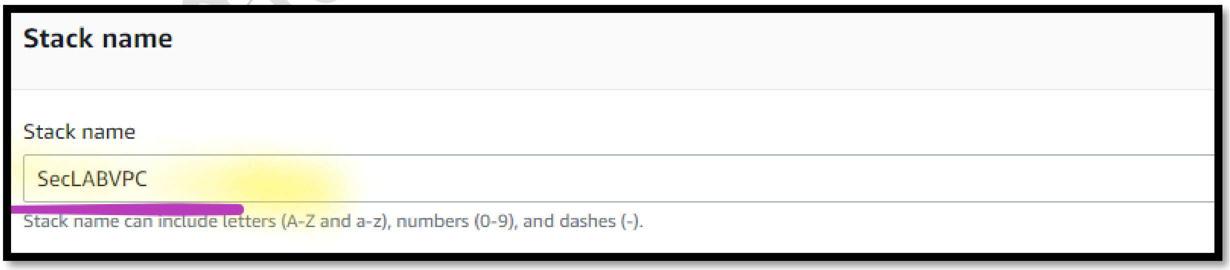
**Note**: **LAB-VPC-01**.yaml template is provided with the lab manual.




iii. Click **Next**.

c. In the **Stack name** section:

i. **Stack name**: Write **LABVPC-01**.




d. In the **Parameters** section:

i. **Environment name**: Write **LAB VPC-01**.

**Note**: Leave other values as default.

ii. Click **Next**.

e. In the **Configure stack options** page:

i. Select **Add Tag**:

i. **Key**: Write **Name**.

ii. **Value**: Write **LAB VPC-01**.

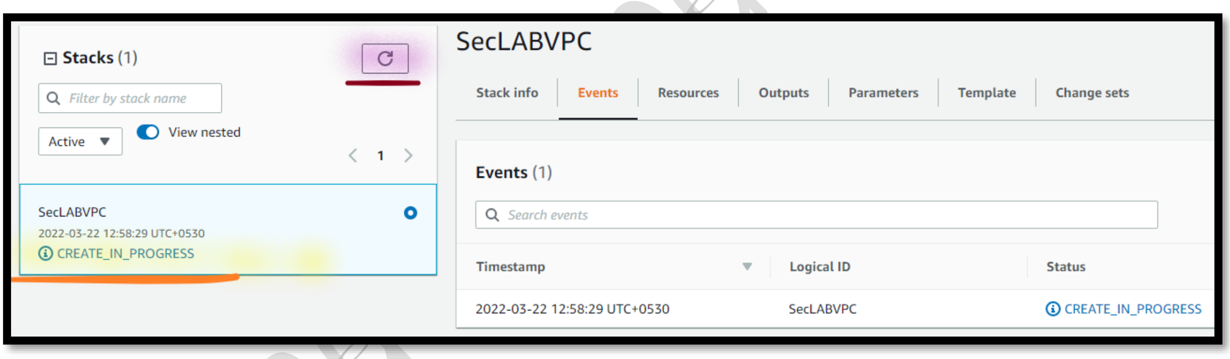
**Note**: Leave all the details as default.

ii. Click **Next**.

f. In the **Review LABVPC01** page:

i. Click **Create stack**.

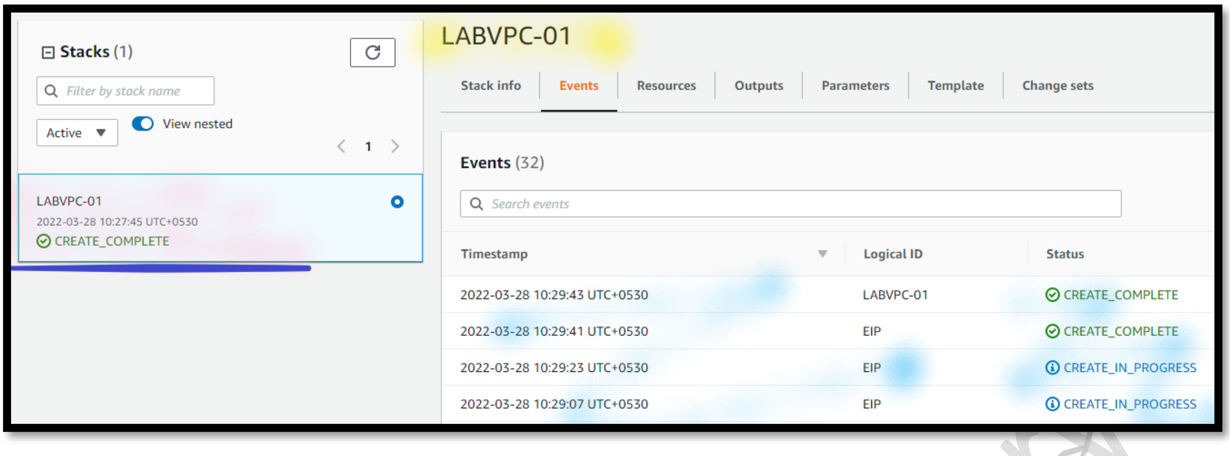
ii. The stack status (***left site under Stack***) is showing as **CREATE_IN_PROGRESS**.




**Note**: Click on **Refresh** until status is turned into **CREATE_COMPLETE**.

**Note**:You need to wait till you can see the **CREATE_COMPLETE** status **under Stack** on your **right site**, as shown below.

**Note**: **Ignore** if you can see any resource **CREATE_IN_PROGRESS** under **Events.**




4. **From** the **LABVPC-01** Stack:

    a. **Go to right**, click the **Outputs** tab.

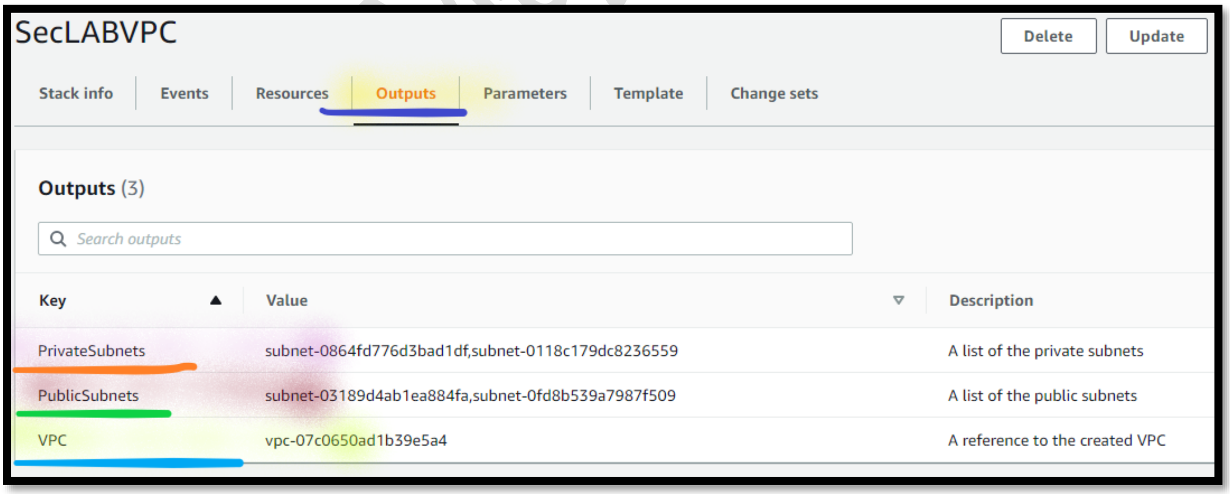
    b. A ***CloudFormation stack*** can provide **output information**, such as resources details. **You will see outputs**:

        i. **PrivateSubnets**: The **Id of the Private Subnets** that was created.

        ii. **PublicSubnets**: The **Id of the Public Subnets** that was created.

        iii. **VPC**: The **Id of the VPC** that was created.






## Step 2: Create Ubuntu Instance

5. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

6. Click **Launch Instance**.

a. In the **Choose an Amazon Machine Image (AMI)** *page*:

   i. Select the **Ubuntu Server 18.04 LTS**.

   ii. Click **Select**.

b. In the **Choose an Instance Type** *page*:

   i. Select the **t2.micro**.

   ii. Click **Next: Configure Instance Details**.

c. In the **Configure Instance Details** *page*:

   i. **Network**: Dropdown and select **LAB VPC-01**.

   ii. **Subnet**: Dropdown and select **Public Subnet-1**.

   iii. **User data**: Select **As file**.

      a) Click on **Choose file** and **navigate** and select **Script_WebApplication**.txt script file.

**Note**: **Script_WebApplication**.txt is available with the Lab manual.

**Note**: Leave the other settings as default.

   iv. Select **Next: Add Storage**.

d. In the **Add Storage** *page:*

**Note**: Leave all the settings as default.

   i. Select **Next: Add Tags**.

e. In the **Add Tags** ***page:***

   i. Click on **Add Tag**:

      a) **Key**: Write **Name**.

      b) **Value**: Write **WebApp Server-01**.

   ii. Select **Next: Configure Security Group**.

f. In the **Configure Security Group** *page:*

i. **Assign a security group**: Select **Create a new security group**.

a) **Security group name**: Write **WebServer-SG**.

b) **Description**: Write **Web Server Security Group**.

**Info**: **SSH Port [22]**, already added and allowed from anywhere [0.0.0.0/0].

c) Select **Add Rule**.

1) **Type**: Dropdown and Select **HTTP**.

2) **Source**: Dropdown and Select **Anywhere**.

**Note**: Leave the other settings as default.

ii. Select **Review and Launch**.

g. In the **Review Instance Launch** *page:*

i. Select **Launch**.

a) In the **Select an existing key pair or create a new key pair** *section:*

1) **First selection box:** Dropdown and select **Choose an existing key pair**.

2) **Key pair name**: Dropdown and select **My-SEC-LAB-KP**.

3) **Enable** the **Checkmark** against the ***I acknowledge that I have access ...***.

**Note**: Leave the other settings as default.

ii. Click on **Launch Instances**.

h. Select the **View Instances**.

### Step 3: Check the Web Server Status

7. **From** the **EC2** console:
8. Click **Instance**.
    a. Select **Web Server**.
        i. **Wait** for the **Instance State** to change to **Running** state.
        ii. **Wait** for the **Status check** to change to **2/2 checks passed**.

**Note**: **Wait** (***~5-10 mnts.***) for the server deployment.

b. Select **Web Server**.
    i. ***Go below*** in the console and click on **Details**.
    ii. **Expand** the ***Instance summary***.
        a) **Copy** the **Public IP address** in the **Notepad**.

## Task 2: Create 2nd VPC with Database

### Step 1: Deploy a 2nd VPC with MySQL Database

AWS CloudFormation model and provision AWS resources in your cloud environment in automated way.

**Info**: In this task, CloudFormation provision 2nd virtual network with two Subnet and create DB Subnet Group and deploy MySQL database in the 2nd virtual network .

5. **From** the **CloudFormation** console.
6. Click **Create stack**.
7. Select **With new resources (standard)**:
    a. In the **Prerequisite - Prepare template** section:
        i. **Prepare template**: Select **Template is ready**.
    b. In the **Specify template** section:

i. **Template source**: Select **Upload a template file**.

ii. **Upload a template file**: Click on **choose file** then select the **Shared-VPC**.yaml file.

**Note**: **Shared-VPC.yaml** template is provided with the lab manual.

iii. Click **Next**.

c. In the **Stack name** section:

i. **Stack name**: Write **SharedVPC**.

d. In the **Parameters** section:

**Note**: Leave other values as default.

i. Click **Next**.

e. In the **Configure stack options** page:

i. Select **Add Tag**:

a) **Key**: Write **Name**.

b) **Value**: Write **Shared VPC**.

**Note**: Leave all the details as default.

ii. Click **Next**.

f. In the **Review SharedVPC** page:

i. Click **Create stack**.

ii. The stack status (***left site under Stack***) is showing as **CREATE_IN_PROGRESS**.

**Note**: Click on **Refresh** until status is turned into **CREATE_COMPLETE**.

**Note**:You need to wait till you can see the **CREATE_COMPLETE** status **under Stack** on your **right site**, as shown below.

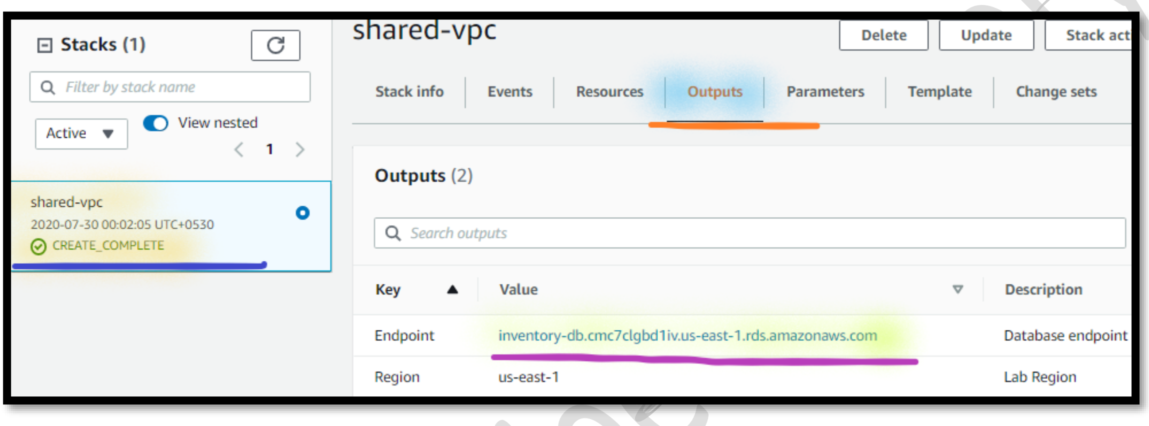
**Note**: **Ignore** if you can see any resource **CREATE_IN_PROGRESS** under **Events.**

8. **From** the **SharedVPC** Stack:

    a. **Go to right**, click the **Outputs** tab.

    b. A ***CloudFormation stack*** can provide **output information**, such as resources details. **You will see outputs**:

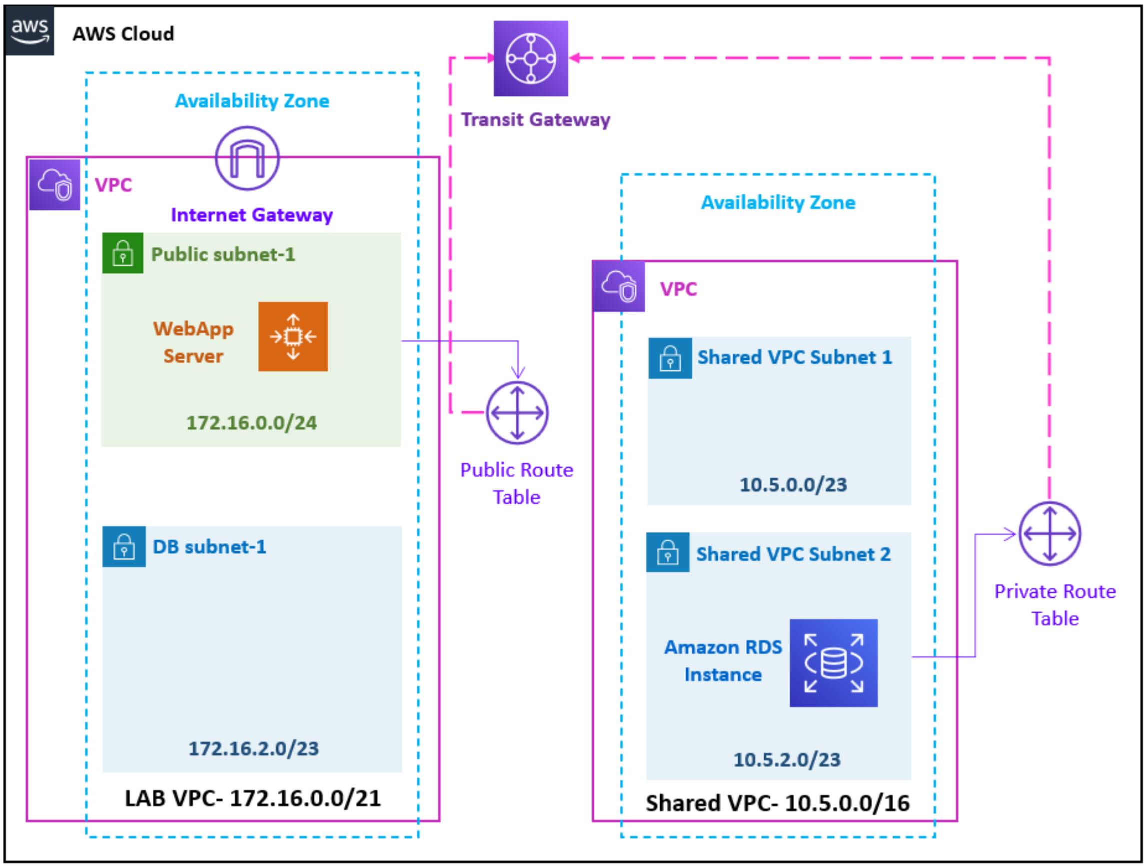
        i. **Endpoint**: **Copy** the **Database endpoint** in the **Notepad**.




## Task 3: Create Transit Gateway for Region 1

AWS Transit Gateway connects VPCs and on-premises networks through a central hub. This simplifies your network and puts an end to complex peering relationships. It acts as a cloud router.

In addition to the regular VPC Routing Table (*used by both Amazon VPC peering and AWS Transit Gateway*); AWS Transit Gateway has its own Routing Table which provides an extra layer of routing flexibility. AWS Transit Gateway Route Tables enables network segmentation by creating multiple route tables and associate Amazon VPCs and other network connections, for example VPNs and Amazon Direct Connect to them.

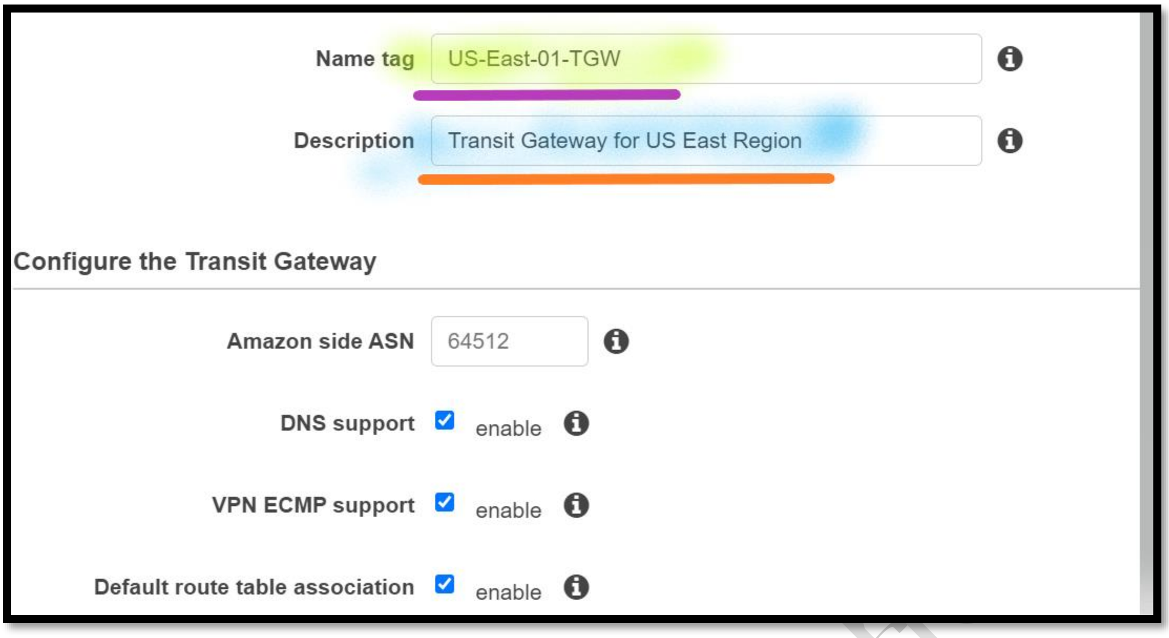



## Step 1: Create Transit Gateway in Region-1

An AWS Transit Gateway is a network transit hub that you can use to interconnect your VPCs and on-premises networks. In this task, you will connect the 3 VPCs thru AWS Transit Gateway.

9. In the **AWS Management Console**, on the **Services** menu, click **VPC**.
10. Choose the **US East (North Virginia)** region list to the right of your account information on the navigation bar.
11. ***Go to the left*** navigation pane, click **Transit Gateway**.
12. Click **Create Transit Gateway** and configure:
    a. **Name**: Write **US-East-01-TGW**.
    b. **Description**: Write **Transit Gateway for US East Region**.

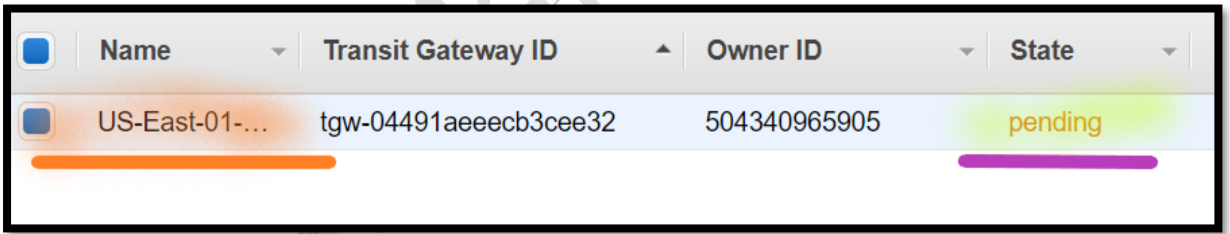
**Note**: Leave other details as default.



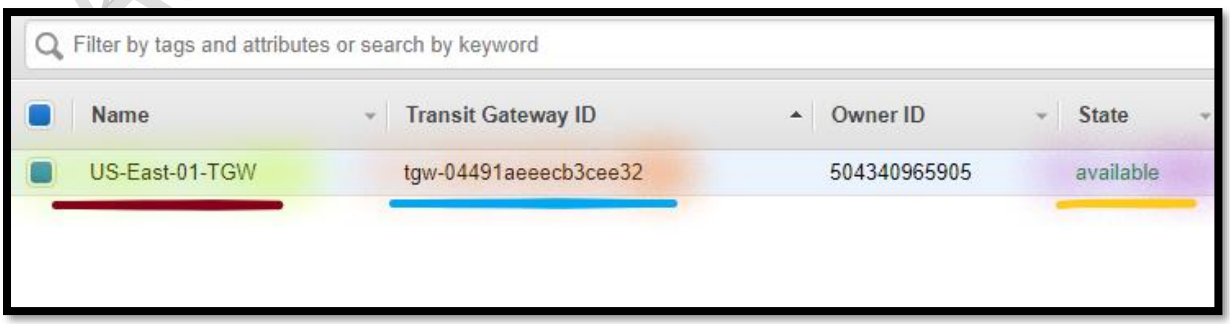

c. Click **Create Transit Gateway**.

d. Select **Close**.

**Note:** You can see the **Transit Gateway** status as **Pending**.




**Note:** **Wait** (**~5 mnts.**), till you can see the **Transit Gateway** status as **Available**. Keep **Refresh**, until you can see status as Available.



**Note: Copy the Transit Gateway Id in Notepad.**

### Step 2: Create Transit Gateway Attachment for LAB VPC

A *transit gateway attachment* is both a source and a destination of packets. You can attach VPCs, VPNs, AWS Direct Connect, or Transit Gateway peering connections. This lab utilizes two types of attachment, namely:

- VPC attachment
- Transit gateway peering attachment

The *VPC attachment* enables the transit gateway to route packets to and from a VPC. While the *Transit Gateway peering attachment* enables routing between transit gateways. A Transit Gateway peering connection attachment is required when routing traffic across multiple Regions using transit gateway.

13. **From** the **VPC** console:

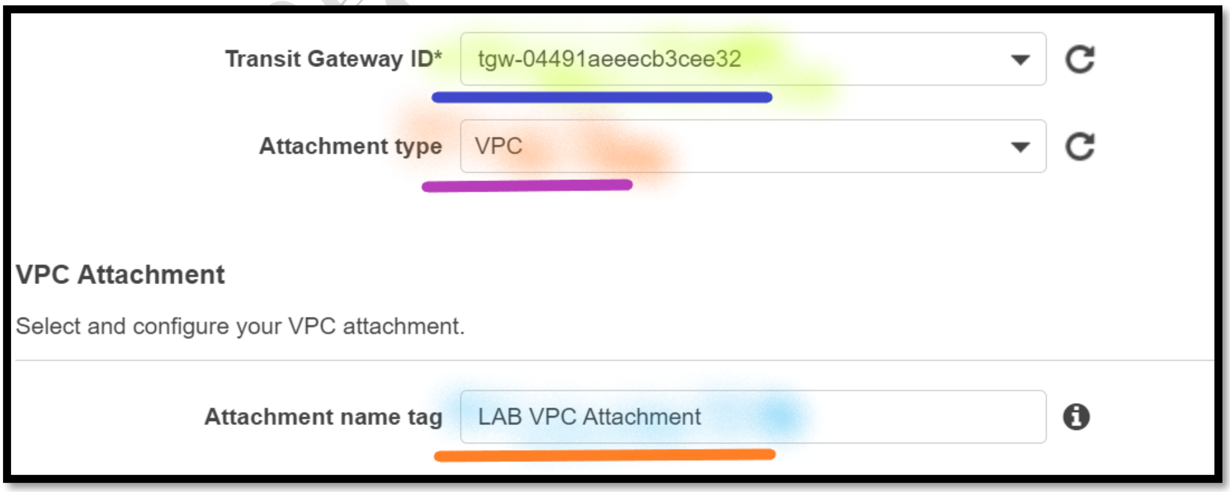
14. ***Go to the left***, select **Transit Gateway Attachment**.

15. Click **Create Transit Gateway Attachment** and configure:

    a. **Attachment tag**: Write **LAB VPC-01 Attachment**.

    b. **Transit Gateway ID**: Dropdown and Select **US-East-01-TGW**.

    c. **Attachment type**: Dropdown and Select **VPC**.




    d. **VPC ID**: Dropdown and Select **LABVPC-01**.

    e. **Availability Zone**: Select **First Availability Zone**.

i. **Subnet ID**: Dropdown and Select **Public Subnet-1**.

f. **Availability Zone**: Select **Second Availability Zone**.

i. **Subnet ID**: Dropdown and Select **Public Subnet-2**.

**Note**: Leave other details as default.


g. Select **Create transit gate attachment**.

## Step 3: Create Transit Gateway Attachment for Shared VPC

16. **From** the **VPC** console:

17. ***Go to the left***, select **Transit Gateway Attachment**.

18. Click **Create Transit Gateway Attachment** and configure:

a. **Attachment tag**: Write **Shared VPC Attachment**.

b. **Transit Gateway ID**: Dropdown and Select **US-East-01-TGW**.

c. **Attachment type**: Dropdown and Select **VPC**.

d. **VPC ID**: Dropdown and Select **Shared VPC**.

e. **Availability Zone**: Select **First Availability Zone**.

i. **Subnet ID**: Dropdown and Select **Shared VPC Subnet 1**.

f. **Availability Zone**: Select **Second Availability Zone**.

i. **Subnet ID**: Dropdown and Select **Shared VPC Subnet 2**.

**Note**: Leave other details as default.


g. Select **Create transit gateway Attachment**.

### Step 4: View Transit Gateway Attachment Status

19. **From** the **VPC** console:

20. ***Go to the left***, click **Transit Gateway Attachment**.

**Note: Wait,** till you can see **Both** the **Transit Gateway Attachment Status** as **Available**. Keep **Refresh**, until you can see status as Available.

### Step 5: View the Transit Gateway Route Table

When you attach a VPC to a Transit Gateway, a route entry must be added in the VPC route table for traffic to route through the Transit Gateway.

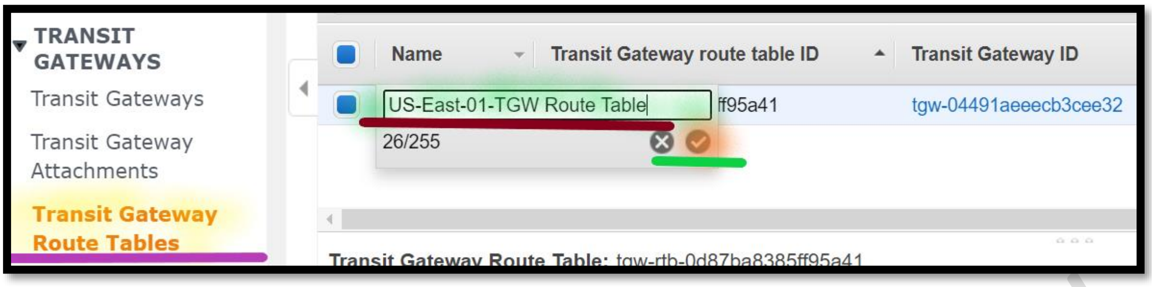
In addition, when you peer Transit Gateways, a static route entry must be added in the Transit Gateway route table to enable traffic to route between the peers.

21. ***Go to the left*** navigation pane, click **Transit Gateway Route Tables**.

a. Select the **Default Transit Gateway Route Tables**.

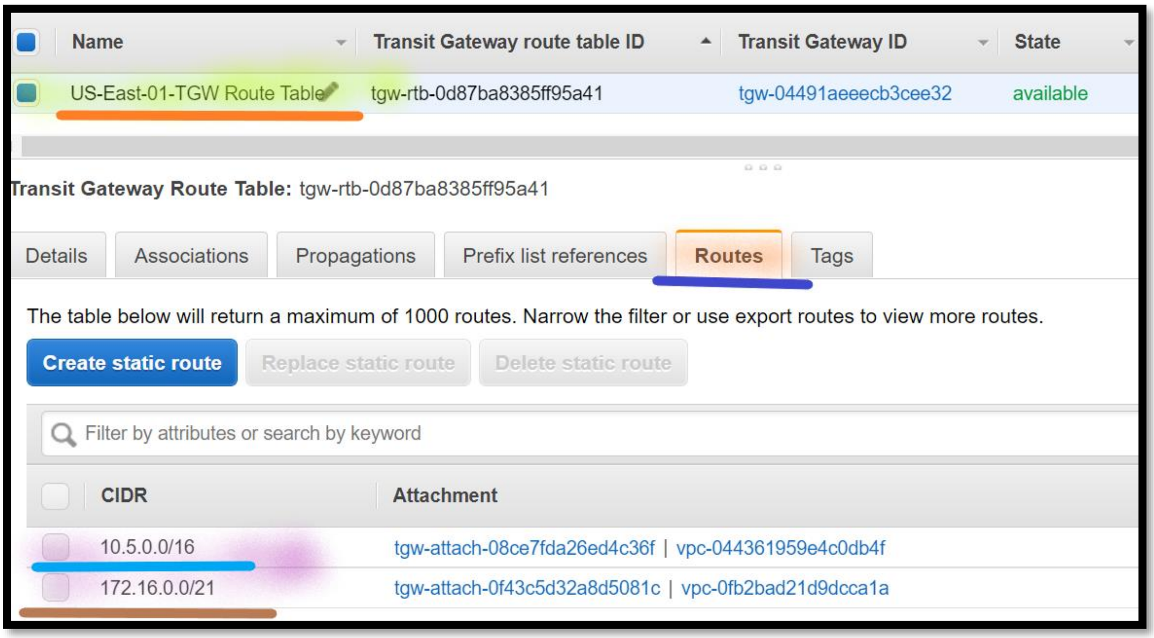
i. **Name**: **Hover** over **Name** section, click on **Pencil Icon** and type **US-East-01-TGW Route Table**.

ii. Click on **Right-click** **icon** to **Save**.




b. ***Go below*** in the Console and Select **Routes**.

**Note**: You can the **Routes** for **Lab VPC-01** [**10.0.0.0/2**] and **Shared VPC** [**10.5.0.0/16**] already added.




c. ***Go below*** in the Console and Select **Associations**.

**Note**: You can see the **Attachment ID** of **Lab VPC** and **Shared VPC**.

d. ***Go below*** in the Console and Select **Propagation**.

**Note**: You can see the **Attachment ID** of **Lab VPC** and **Shared VPC**.

### Step 6: Configure Route Tables attached to LAB VPC

In this section, add the route for Shared VPC.

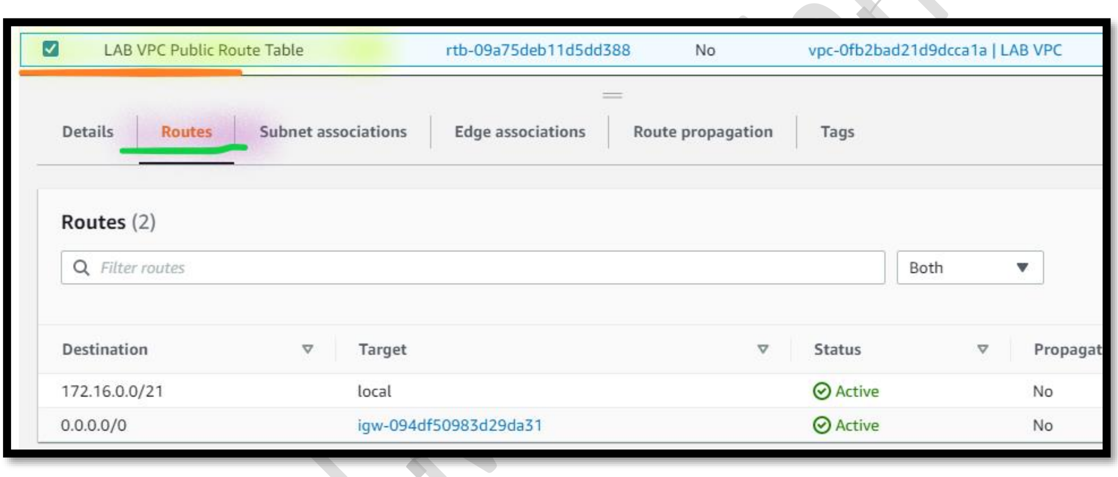
22. ***Go to the left*** navigation pane, Click **Route Tables**.

23. Select **Lab VPC-01_Public Route Table** (for ***LAB VPC***).

    a. ***Go below*** in the console, Select **Routes** tab.

    b. Click **Edit routes** then configure:

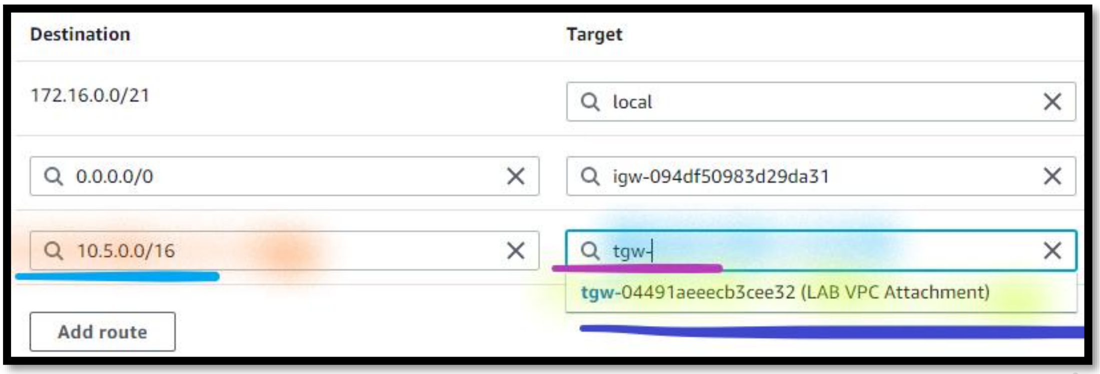
**Note**: If you are seeing any **Route Status** as **Blackhole**, **Remove** the same.




i. Click **Add route**.

ii. **Destination**: Write **10.5.0.0/16** (The is the ***CIDR range of Shared VPC***).

iii. **Target**: **Click** on the **box** in the **Target** and select **Transit Gateway**. Select **LAB VPC Attachment** (which you have created in the previous step)




iv. Click **Save changes**.

> **Info**: You will now configure the reverse flow for traffic coming from ***Shared VPC*** that is going to ***Lab VPC***.

### Step 7: Configure Route Tables attached to Shared VPC

In this section, add the route for LAB VPC.

24. Select **Shared VPC Route Table** (for ***Shared VPC***)

> **Info**: You will configure the *Route Table* associated with ***Shared VPC*** to send traffic to the peering connection if the destination IP address falls within the range of the *Lab VPC*.

a. ***Go below*** in the console, Select **Routes** tab.

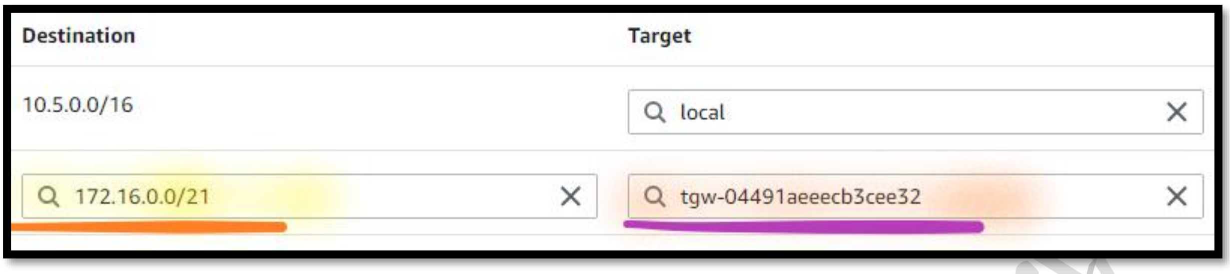
b. Click **Edit routes** then configure:

> **Note**: If you are seeing any **Route Status** as **Blackhole**, **Remove** the same.

i. Click **Add route**.

ii. **Destination**: Write **172.16.0.0/21** (The is the ***CIDR range of LAB VPC***).

v. **Target**: **Click** on the **box** in the **Target** and select **Transit Gateway**. Select **Shared VPC Attachment** (which you have created in the previous step)



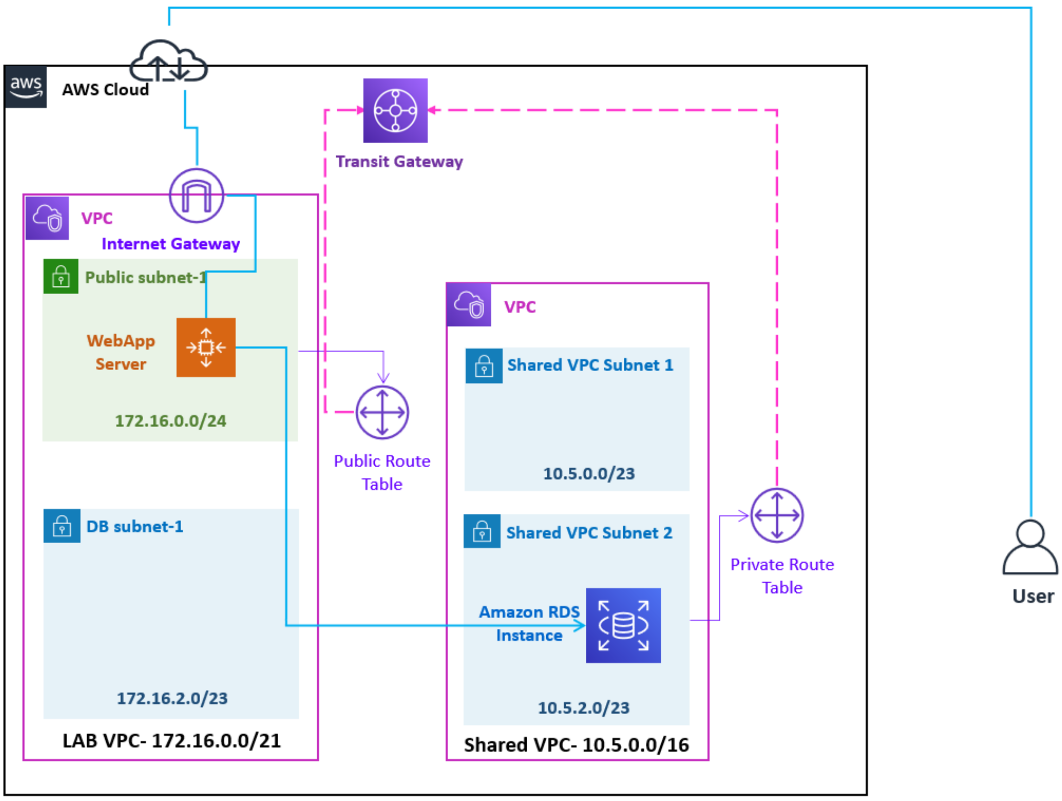
| Destination | Target |
|---|---|
| 10.5.0.0/16 | local |
| 172.16.0.0/21 | tgw-04491aeeecb3cee32 |

iii. Click **Save changes**.

**Info**: The route tables have now been configured to send traffic via the peering connection when the traffic is destined for the other VPC.

## Task 4: Test the Connectivity

In this section, test the connectivity between Lab VPC and Shared VPC.



## Step 1: Connect to Web Server

25.**From** the **Local Desktop/ Laptop**, right click on **Start** & **Run**.

a. **Connect** to **WebApp-Server-01** **using** the **putty**.

**Note**: **Map** your ***.ppk** file before connecting to **Web Server-01** via **putty**.

i. **Login as**: Write username as **ubuntu**.

## Step 2: Test the Connectivity

26.**From** the **WebApp-Server-01** instance **console**, ***Get*** the **MySQL RDS** Instance **Private IP address**:

ping **HOSTNAME**

**Note**: **Replace** the **HOSTNAME** with **RDS MySQL instance endpoint**.

**Note**: **Copy** the **RDS MySQL instance Private IP address** in **Notepad**.

**Note**: Press **CTRL + C** to exit.

```
ubuntu@ip-172-16-0-17:~$
ubuntu@ip-172-16-0-17:~$ ping inventory-db.cmc7clgbdliv.us-east-1.rds.amazonaws.com
PING inventory-db.cmc7clgbdliv.us-east-1.rds.amazonaws.com (10.5.3.21) 56(84) bytes of data.

^C
--- inventory-db.cmc7clgbdliv.us-east-1.rds.amazonaws.com ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2049ms

ubuntu@ip-172-16-0-17:~$
```

27.**From** the **WebApp-Server-01** instance **console**, ***Test*** the **MySQL RDS** Instance **connectivity**:

nc -zvw3 **RDS-Private-IP-address** 3306

**Note**: **Replace** the **RDS-Private-IP-address** with **Amazon RDS MySQL instance Private IP address**.

**Info**: Netcat tool use for port scanning and port listening.

```
ubuntu@ip-172-16-0-17:~$
ubuntu@ip-172-16-0-17:~$ nc -zvw3 inventory-db.cmc7clgbdliv.us-east-1.rds.amazonaws.com 3306
Connection to inventory-db.cmc7clgbdliv.us-east-1.rds.amazonaws.com 3306 port [tcp/mysql] succeeded!
ubuntu@ip-172-16-0-17:~$
```

**Note**: In the **Output** you can see the **Succeeded** message. It means **connectivity is established** between **LAB VPC** and **Shared VPC**.

### Step 3: Create Table in Prod_Schema Database

You will now configure the MySQL Database.

28. **From** the **WebApp-Server-01** instance **console**, ***Connect*** to MySQL RDS Instance.
    mysql -u master -p -h **HOSTNAME**

**Note**: **Replace** the **HOSTNAME** with **Amazon RDS MySQL instance endpoint**.

29. You will be **prompted** to enter the password. Enter **lab-password**.
    a. You will be shown a ***brief introduction message*** and then be placed at the **mysql>** prompt.

    b. **List** the ***existing databases***.
       show databases;

    c. **Use** the ***Inventory database***.
       use prod_schema;

d. **Create** **Products table** in the ***Inventory database***.

```
create table `products` (
  `id` int(11) not null auto_increment,
  `name` varchar(45) not null,
  `quantity` varchar(45) not null,
  `price` varchar(45) not null,
  primary key (`id`),
  unique key `id_unique` (`id`));
```

e. **List** the tables in the ***Inventory database***.

```
show tables;
```

## Step 4: Create Database Connection String

30. **Update** the below Connection string in **Notepad**

Server=**HOST-NAME**;uid=master;pwd=lab-password

**Note**: **Replace** the **HOSTNAME** with **Amazon RDS MySQL instance endpoint**.

## Step 5: Access WebApp Server

31. **From** your **local desktop/ laptop** **Web browser**, type **Public IP Address** of **WebApp-Server-01**.

**Note**: You can also see **Private IP address** of the **WebApp-Server** in the **Web Page**.

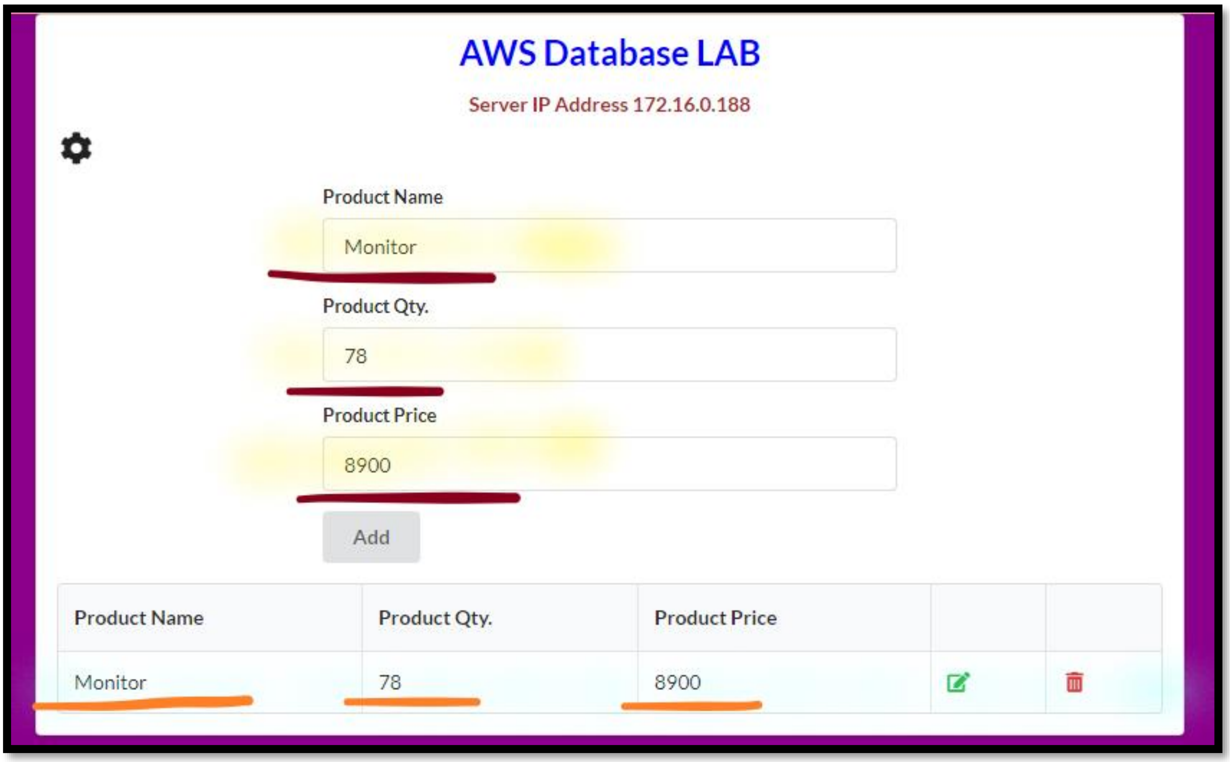
a. Open the **Settings** (**gear icon** in the web page).

b. You get the **Prompt** to enter the **Connection String**.

   i. **Type** **Connection string** (which you have created in the previous step).

   ii. Select **Ok**.

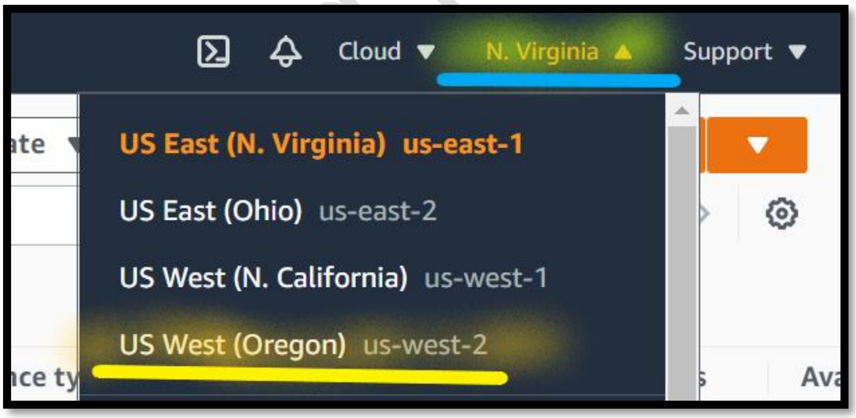
c. **Add** **Product inventory** details.




## Task 5: Create VPC with Web Server

### Step 1: Create New Key Pair

32.Choose the **US West (Oregon)** region list to the right of your account information on the navigation bar.




33.In the AWS Management Console, on the **Services** menu, click **EC2**.

34.***Go to left***, Select **Key Pairs**.

a. Select **Create key pair**.

i. **Name**: Write **My-SEC-LAB-KP-R2**.

ii. **Key pair type**: Select RSA.

iii. **Private key file format**: Select .ppk.

iv. Select Create Key Pair.

**Note**: My-SEC-Lab-KP-R2.ppk, downloaded in your local desktop/ laptop. Keep it secure.

### Step 2: Deploy a VPC using CloudFormation

35. In the **AWS Management Console**, on the Services menu, click CloudFormation.

36. Dropdown and select the US East (N. Virginia) region.

37. Click Create stack:

g. In the Prerequisite - Prepare template section:

i. **Prepare template**: Select Template is ready.

h. In the Specify template section:

i. **Template source**: Select Upload a template file.

ii. **Upload a template file**: Click on choose file then select the LAB-VPC-02.yaml file.

**Note**: LAB-VPC-02.yaml template is provided with the lab manual.

iii. Click Next.

i. In the Stack name section:

i. **Stack name**: Write LABVPC-02-R2.

j. In the Parameters section:

**Note**: Leave other values as default.

i. Click Next.

k. In the Configure stack options page:

i. Select Add Tag:

i. **Key**: Write **Name**.

ii. **Value**: Write **LAB VPC-02-R2**.

**Note**: Leave all the details as default.

ii. Click **Next**.

l. In the **Review LABVPC02-R2** page:

i. Click **Create stack**.

ii. The stack status (***left site under Stack***) is showing as **CREATE_IN_PROGRESS**.

**Note**: Click on **Refresh** until status is turned into **CREATE_COMPLETE**.

**Note**:You need to wait till you can see the **CREATE_COMPLETE** status **under Stack** on your **right site**, as shown below.

**Note**: **Ignore** if you can see any resource **CREATE_IN_PROGRESS** under **Events**.

38.**From** the **LABVPC-02-R2** Stack:

c. **Go to right**, click the **Outputs** tab.

d. A ***CloudFormation stack*** can provide **output information**, such as resources details. **You will see outputs**:

i. **Public IP**: The **Public IP address** of ***Web Server***.

**Note**: **Copy** the **Public IP address** in **Notepad**.

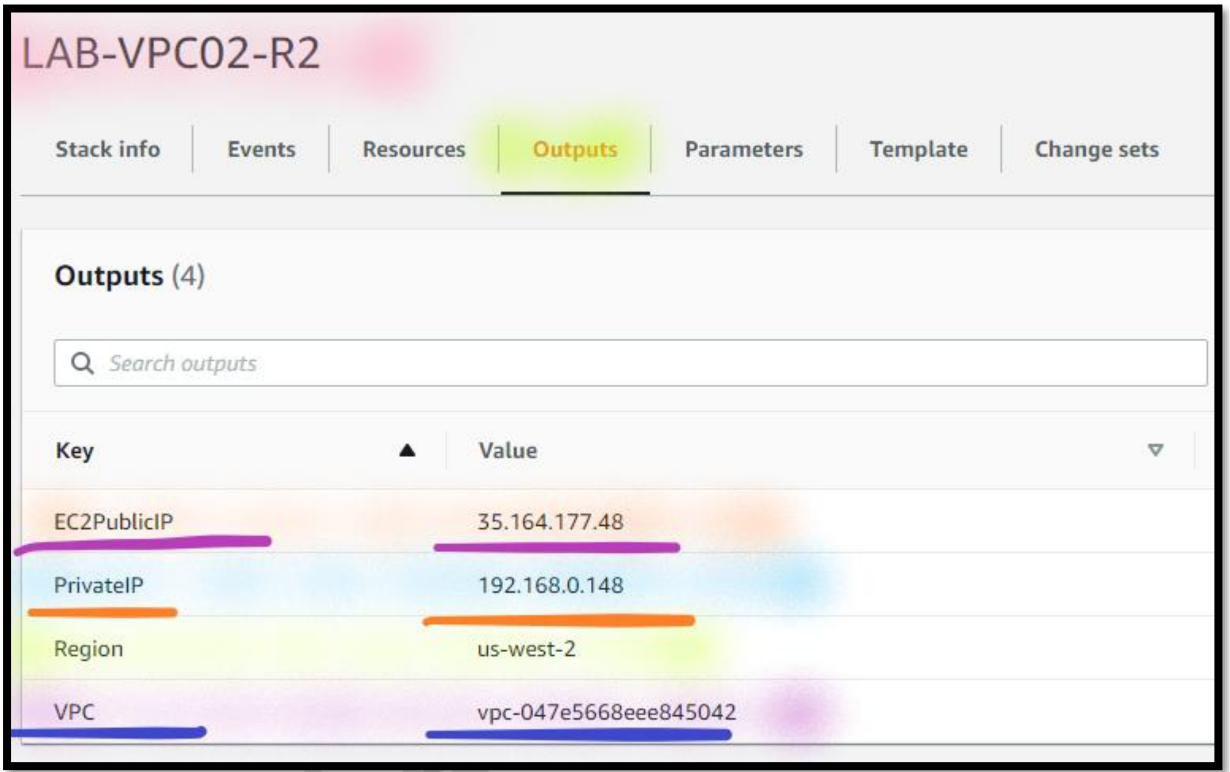
ii. **Private IP**: The **Private IP address** of ***Web Server***.

**Note**: **Copy** the **Private IP address** in **Notepad**.

iii. **Region**: The **Region**, where **Lab-VPC03** and **EC2 Instance** deployed.

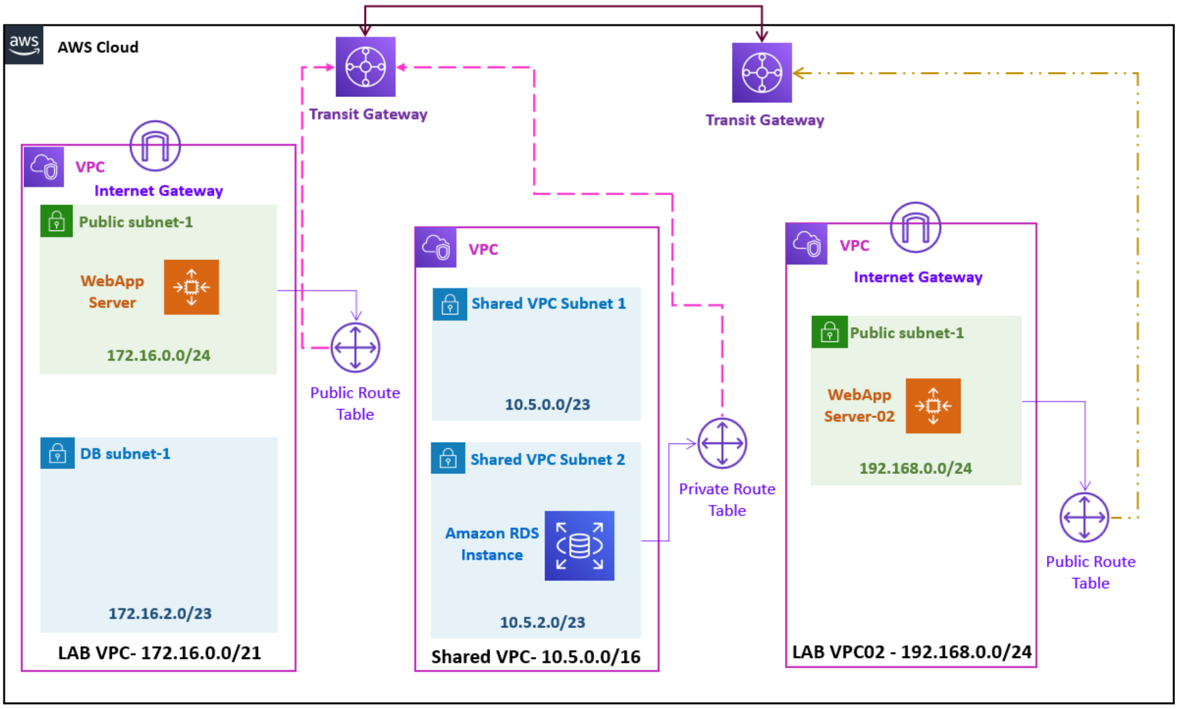
iv. **VPC**: The **VPC Id** of ***LAB VPC02_R2***.

**Note**: **Copy** the **VPC Id** in **Notepad**.



# Task 4: Create Transit Gateway for Region 2

In this section, create a transit gateway for US-West-2 region.




## Step 1: Connect to Web Server

39.From the **Local Desktop/ Laptop** (Windows), right click on **Start** & **Run**.

a. **Connect** to **WebApp-Server-02-R2** **using** the **putty**.

> **Note**: **Map** your ***.ppk** file before connecting to **WebApp-Server-02-R2** via **putty**.

i. **Login as**: Write username as **ubuntu**.

## Step 2: Test the Connectivity

In this section, test the connectivity between a Lab VPC02 and Lab VPC and Lab VPC02 and Shared VPC.

40.**From** the **WebApp-Server-02-R2** instance **console**, ***Test*** the **MySQL RDS** Instance ***connectivity***:

nc -zvw3 **RDS-Private-IP-address** 3306

**Note**: **Replace** the **RDS-Private-IP-address** with **Amazon RDS MySQL instance Private IP address**.

**Note**: In the **Output** you can see the **Timed out** message. It means there is **no connectivity established** between **LAB VPC02-R2** and **Shared VPC**.

41. **From** the **WebApp-Server-02_R2** instance **console**, ***Test*** the **WebApp-Server-01** Instance ***connectivity***:

    nc -zvw3 **WebApp-Server01-Private-IP-Address** 80

**Note**: **Replace** the **WebApp-Server01-Private-IP-Address** with **WebApp-Server-01-Private-IP-Address** (which you have copied in the previous step).

**Note**: In the **Output** you can see the **Timed out** message. . It means there is **no connectivity established** between **LAB VPC02-R2** and **LAB VPC**.

### Step 3: Create Transit Gateway in Region-2

In this section, create a transit gateway for West-US-2 region.

42. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

43. Choose the **US West (Oregon)** region list to the right of your account information on the navigation bar.

44. ***Go to the left*** navigation pane, click **Transit Gateway**.

45. Click **Create Transit Gateway** and configure:

    a. **Name**: Write **US-West-02-TGW**.

    b. **Description**: Write **Transit Gateway for US West Region**.

**Note**: Leave other details as default.

    c. Click **Create Transit Gateway**.

d. Select **Close**.

**Note:** You can see the **Transit Gateway** status as **Pending**.

**Note:** **Wait**, till you can see the **Transit Gateway** status as **Available**. Keep **Refresh**, until you can see status as Available.

## Step 5: Create Transit Gateway Attachment for LAB VPC02-R2

In this section, create a transit gateway attachment for VPC.

46. **From** the **VPC** console:

47. ***Go to the left***, select **Transit Gateway Attachment**.

48. Click **Create Transit Gateway Attachment** and configure:

    a. **Attachment tag**: Write **LAB VPC02_R2 Attachment**.

    b. **Transit Gateway ID**: Dropdown and Select **US-West-02-TGW**.

    c. **Attachment type**: Dropdown and Select **VPC**.

    d. **VPC ID**: Dropdown and Select **LAB VPC02-R2**.

    e. **Availability Zone**: Select **First Availability Zone**.

        i. **Subnet ID**: Dropdown and Select **LAB VPC02-R2-Public-Subnet**.

**Note**: Leave other details as default.

    f. Select **Create transit gateway attachment**.

## Step 6: Create Transit Gateway Attachment for Region-1 Peering

In this section, create a transit gateway attachment for Peering.

49. **From** the **VPC** console:

50. ***Go to the left***, select **Transit Gateway Attachment**.

51. Click **Create Transit Gateway Attachment** and configure:

   a. **Attachment tag**: Write **Peering-with-EastUS-Attachment**.

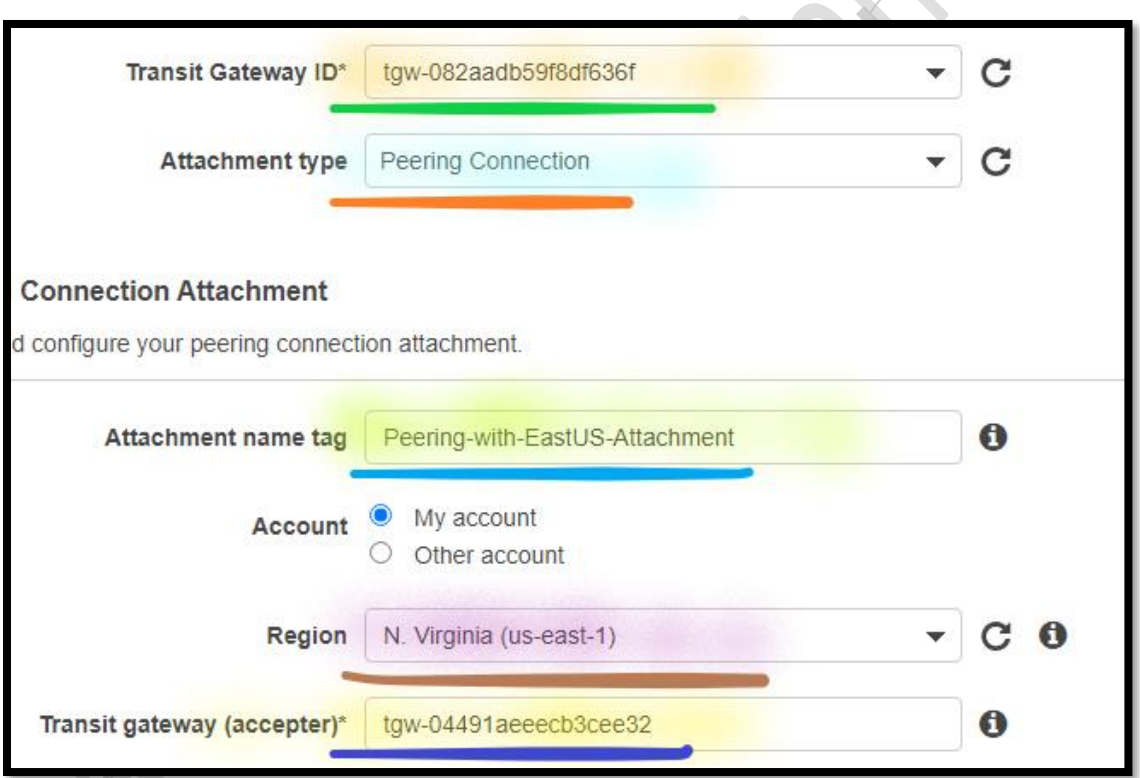
   b. **Transit Gateway ID**: Dropdown and Select **US-West-02-TGW**.

   c. **Attachment type**: Dropdown and Select **Peering Connection**.

   d. **Region**: Dropdown and Select **N. Virginia (US-East-1)**.

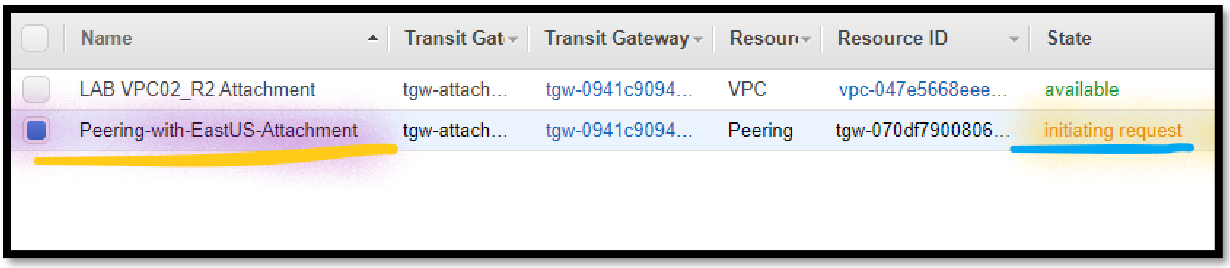
   e. **Transit gateway (accepter)**: Write **US-East-01-TGW ID** (which you have copied in the previous step).

**Note**: Leave other details as default.




   f. Select **Create transit gateway attachment**.

**Note:** You can see the **Transit Gateway Attachment** state as **initiating request**.




## Step 7: Accept the Transit Gateway Attachment from Region-1

**Note**: **Return** to the **Browser tab** of the **N. Virginia (US-East-1)** region.

52. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

53. ***Go to the left***, select **Transit Gateway Attachment**.

**Note:** You can see the **Transit Gateway Attachment** state as **initiating request**.

54. Select the New **Transit Gateway Attachment**.

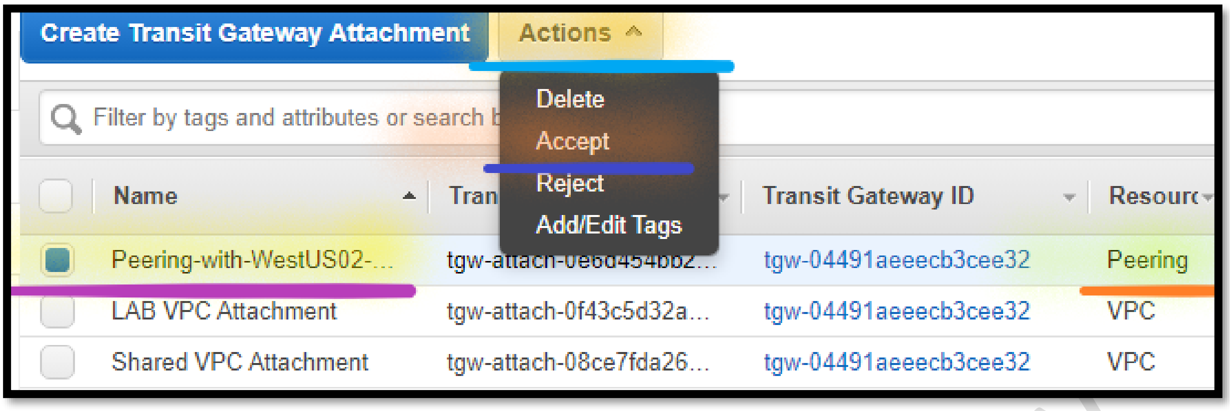
    a. **Name**: **Hover** over **Name** section click on **Pencil Icon** and type **Peering-with-WestUS02-Attachment**.

    b. Click on **Right-click icon** to **Save**.

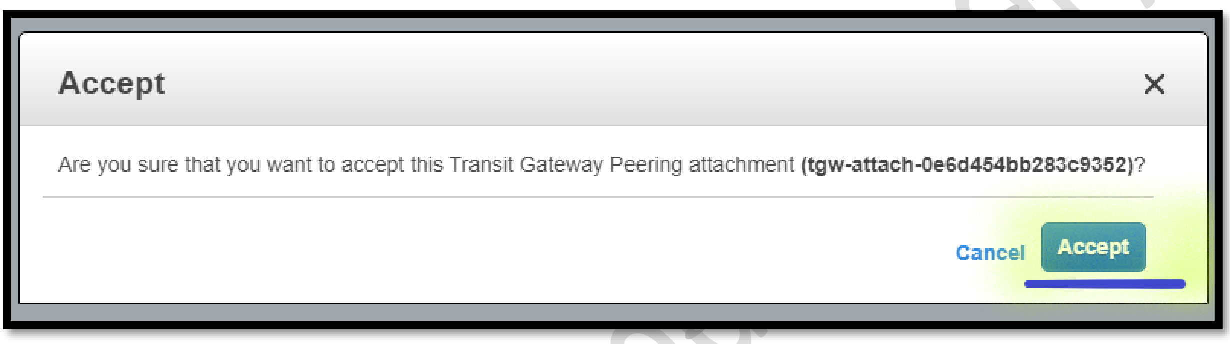

55. Select the **Peering-with-WestUS02-Attachment**.

    a. Select **Action**.

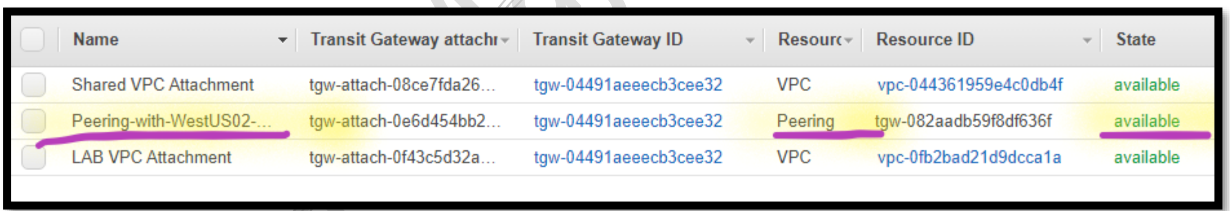
    b. Select **Accept transit gateway attachment**.




c. Select **Accept**.




**Note: Wait (~ 5 minutes)**, till you can see **Transit Gateway Attachment** state as **Available**.




## Step 8: View the Transit Gateway Attachment State from Region-2

**Note**: **Return** to the **Browser tab** of the **Oregon (US-West-2)** region.

56. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

57. ***Go to the left***, select **Transit Gateway Attachment**.

**Note:** You can see **Transit Gateway Attachment** state as **Available**.

## Step 9: View the Transit Gateway Route Table in Region-2

58. ***Go to the left*** navigation pane, click **Transit Gateway Route Tables**.

   a. Select the **Default Transit Gateway Route Tables**.

   b. **Name**: **Hover** over **Name** section and click on **Pencil Icon** and type **US-West-02_TGW Route Table** and **Click** on **Right Checkmark** icon to **Save**.

   c. ***Go below*** in the Console and Select **Associations**.

**Note**: You can the **Attachment ID** of **LAB VPC02-R2** and **Peering**.

   d. ***Go below*** in the Console and Select **Propagation**.

**Note**: You can the **Attachment ID** of **LAB VPC02-R2**.

   e. ***Go below*** in the Console and Select **Routes**.

**Note**: You can see the **Routes** for **LAB VPC02-R2** [**192.168.0.0/24**] already added.

## Step 10: Configure Transit Gateway Route Table in Region-2

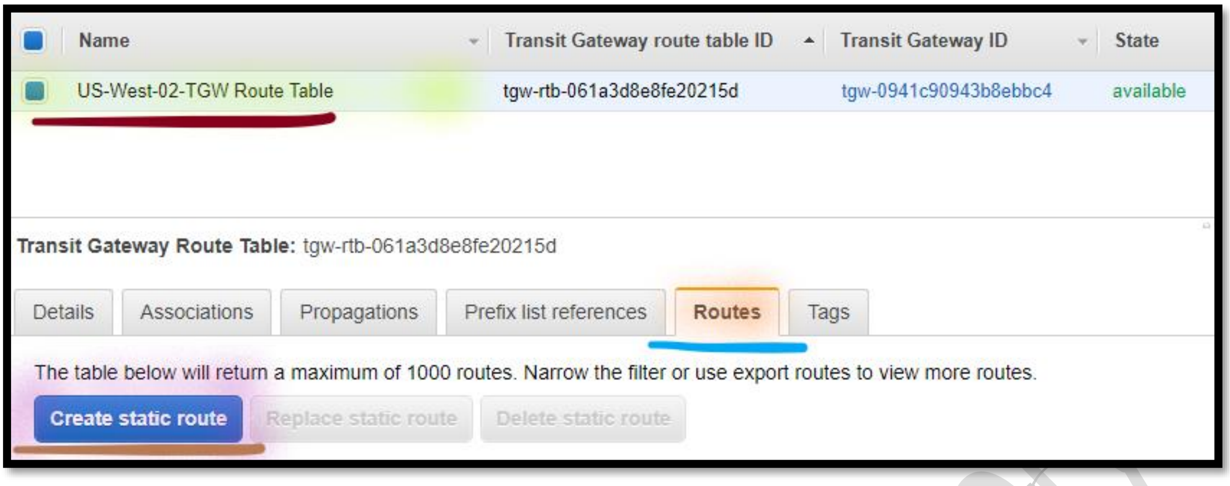
In this section, add the static route of the LAB VPC and Shared VPC in the Transit gateway route table.

59. ***Go to the left*** navigation pane, click **Transit Gateway Route Tables**.

60. Select the **US-West-02-TGW Route Table**.

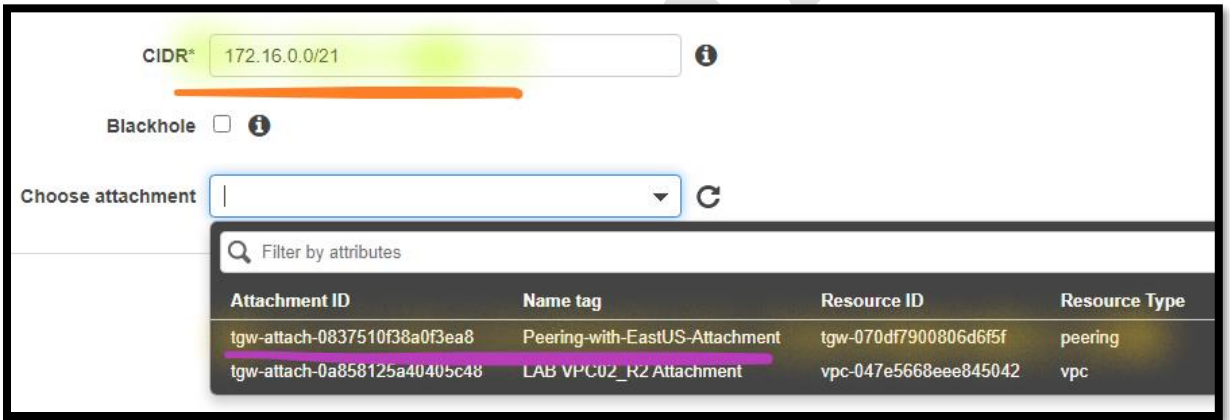
   a. ***Go below*** in the **Console**, Select **Routes**.

   b. Select **Create static route**.




i. **CIDR**: Write **172.16.0.0/21** (The is the ***CIDR range of LAB VPC***).

ii. **Choose attachment**: Dropdown and Select **Peering-with-EastUS-Attachment**.




iii. Select **Create static route**.

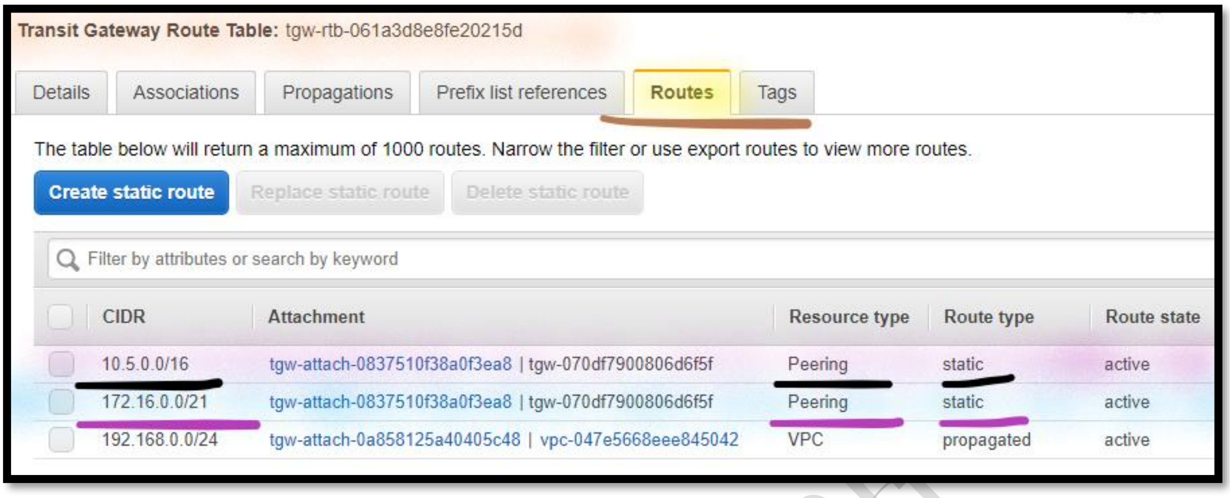
c. Select **Create static route**.

i. **CIDR**: Write **10.5.0.0/16** (The is the ***CIDR range of Shared VPC***).

ii. **Choose attachment**: Dropdown and Select **Peering-with-EastUS-Attachment**.

iii. Select **Create static route**.

**Note**: You can see the **Routes** for **LAB VPC** [**172.16.0.0/21**] and **Shared VPC** [**10.5.0.0/16**] added.




## Step 11: Configure Transit Gateway Route Table in Region-1

In this section, add the static route of the LAB VPC02 in the Transit gateway route table.

**Note**: **Return** to the **Browser tab** of the **N. Virginia (US-East-1)** region.

61. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

62. ***Go to the left*** navigation pane, click **Transit Gateway Route Table**.

63. Select the **US-East-01-TGW Route Table**.

    a. ***Go below*** in the **Console**, Select **Routes**.

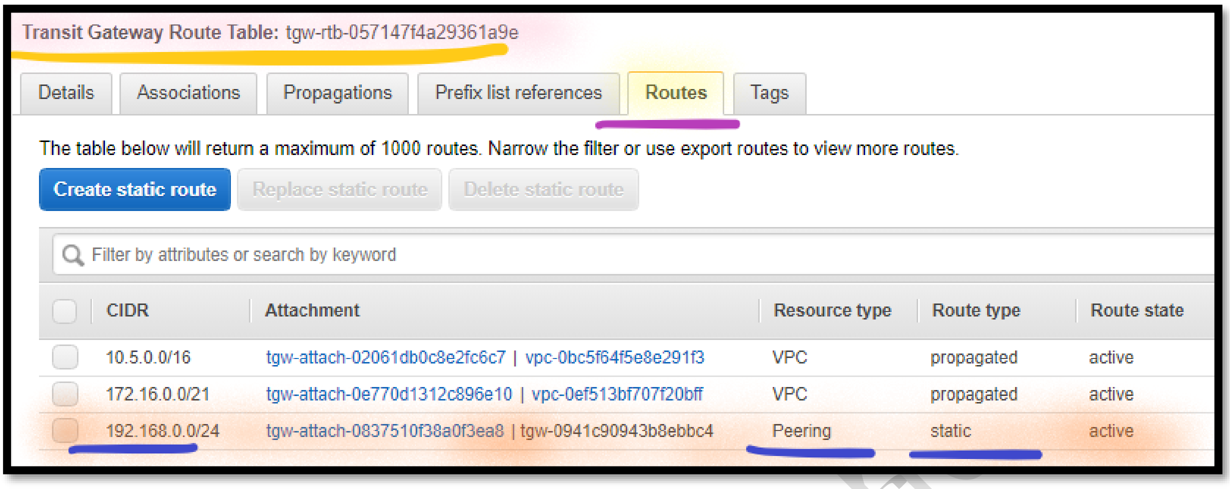
    b. Select **Create static route**.

        i. **CIDR**: Write **192.168.0.0/24** (The is the ***CIDR range of LAB VPC02-R2***).

        ii. **Choose attachment**: Dropdown and Select **Peering-with-WestUS02-Attachment**.

        iii. Select **Create static route**.

**Note**: You can see the **Routes** for **LAB VPC02-R2** [**192.168.0.0/24**] added.




## Step 12: Configure Route Tables attached to Shared VPC

In this section, add the route of the LAB VPC02 in the Shared VPC route table.

64. ***Go to the left***, select **Route tables**.

65. Select **Shared VPC Route Table** (for ***Shared VPC***)

   a. ***Go below*** in the console, Select **Routes** tab.

   b. Click **Edit routes** then configure:

**Note**: If you are seeing any **Route Status** as **Blackhole**, **Remove** the same.

   i. Click **Add route**.

   ii. **Destination**: Write **192.168.0.0/24** (The is the ***CIDR range of LAB VPC02-R2***).

   iii. **Target**: **Click** on the **box** in the **Target** and select **Transit Gateway**. Select **Shared VPC Attachment** (which you have created in the previous step)

   iv. Click **Save changes**.

### Step 13: Configure Route Tables attached to LAB VPC

In this section, add the route of the Lab VPC02 in the Lab VPC route table.

66. Select **LAB VPC-01 Public Route Table** (for ***LAB VPC***)

    a. ***Go below*** in the console, Select **Routes** tab.

    b. Click **Edit routes** then configure:

**Note**: If you are seeing any **Route Status** as **Blackhole**, **Remove** the same.

v. Click **Add route**.

vi. **Destination**: Write **192.168.0.0/24** (The is the ***CIDR range of LAB VPC02-R2***).

vii. **Target**: **Click** on the **box** in the **Target** and select **Transit Gateway**. Select **LAB VPC-01 Attachment** (which you have created in the previous step)

viii. Click **Save changes**.

### Step 14: Configure Route Tables attached to LAB VPC02_R2

In this section, add the route of the Lab VPC and Shared VPC in the Lab VPC02 route table.

**Note**: **Return** to the **Browser tab** of the **Oregon (US-West-2)** region.

67. Select **LAB VPC02-R2 Public Route Table** (for ***LAB VPC02-R2***)

    a. ***Go below*** in the console, Select **Routes** tab.

    b. Click **Edit routes** then configure:

**Note**: If you are seeing any **Route Status** as **Blackhole**, **Remove** the same.

i. Click **Add route**.

ii. **Destination**: Write **10.5.0.0/16** (The is the ***CIDR range of Shared VPC***).

iii. **Target**: **Click** on the **box** in the **Target** and select **Transit Gateway**. Select **LAB-VPC02_R2 Attachment** (which you have created in the previous step).

iv. **Destination**: Write **172.16.0.0/21** (The is the ***CIDR range of LAB VPC***).

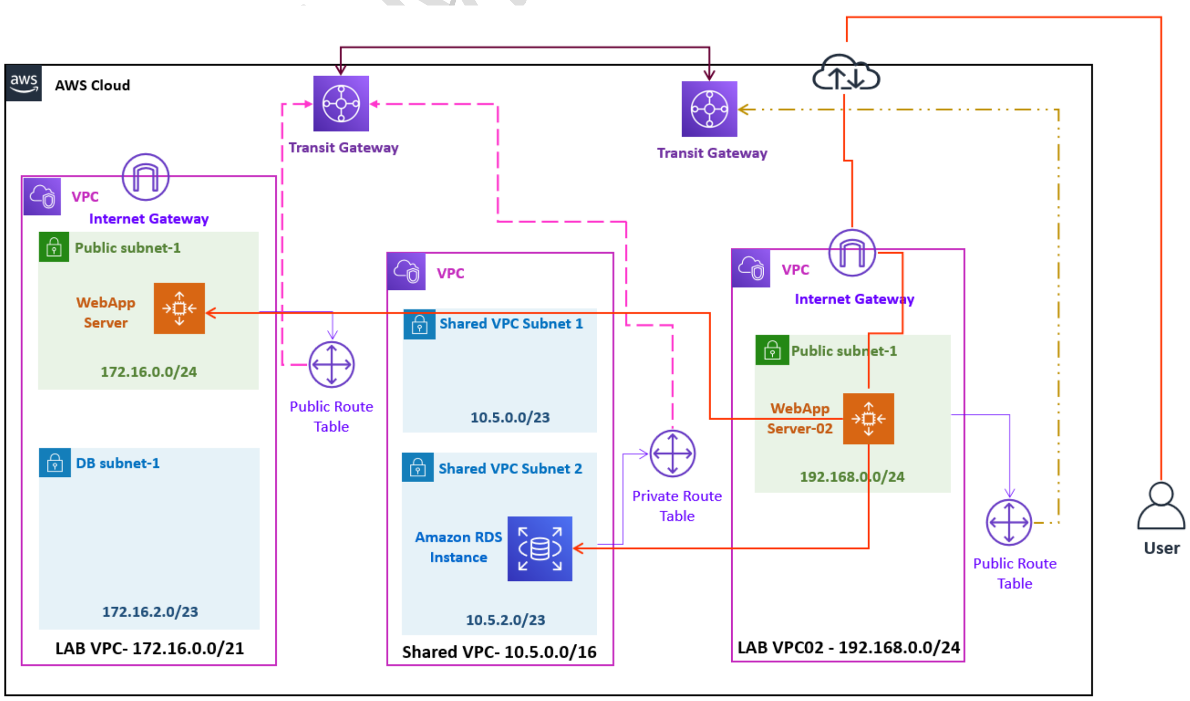
v. **Target**: **Click** on the **box** in the **Target** and select **Transit Gateway**. Select **LAB-VPC02_R2 Attachment** (which you have created in the previous step).

vi. Click **Save changes**.

## Task 4: Test the Connectivity

In this section, test the connectivity between a Lab VPC02 and Lab VPC & Lab VPC02 and Shared VPC.

**Note**: **Return** to the **Browser tab** of the **Oregon (US-West-2)** region.



### Step 1: Connect to Web Server

68. From the **Local Desktop/ Laptop** (Windows), right click on **Start** & **Run**.

    a. **Connect** to **WebApp-Server-02-R2** **using** the **putty**.

**Note**: **Map** your ***.ppk** file before connecting to **WebApp-Server-02-R2** via **putty**.

i. **Login as**: Write username as **ubuntu**.

### Step 2: Test the Connectivity

69. **From** the **WebApp-Server-02-R2** instance **console**, ***Test*** the **MySQL RDS** Instance ***connectivity***:
    nc -zvw3 **RDS-Private-IP-address** 3306

**Note**: **Replace** the **RDS-Private-IP-address** with **Amazon RDS MySQL instance Private IP address**.

**Note**: In the **Output** you can see the **Succeeded** message. It means **connectivity is established** between **LAB VPC02-R2** and **Shared VPC**.

70. **From** the **WebApp-Server-02_R2** instance **console**, ***Test*** the **WebApp-Server-01** Instance ***connectivity***:
    nc -zvw3 **WebApp-Server01-Private-IP-Address** 80

**Note**: **Replace** the **WebApp-Server01-Private-IP-Address** with **WebApp-Server-01-Private-IP-Address** (which you have copied in the previous step).

**Note**: In the **Output** you can see the **Timed out** message. It means **connectivity is established** between **LAB VPC02-R2** and **LAB VPC**.

**Note**: **Replace** the **HOSTNAME** with **Amazon RDS MySQL instance endpoint**.

### Step 3: Access WebApp Server

71. **From** your **local desktop/ laptop** **Web browser**, type **Public IP Address** of **WebApp-Server-02**.

**Note**: You can also see **Private IP address** of the **WebApp-Server** in the **Web Page**.

d. Open the **Settings** (**gear icon** in the web page).

e. You get the **Prompt** to enter the **Connection String**.

i. **Type** **Connection string** (which you have created in the previous step).

ii. Select **Ok**.

f. **Add** **additional** **Product inventory** details.

## Task 5: Clean up the Environment

### Step 1: Delete Transit Gateway and Attachment from Region-1

72. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

73. Dropdown and select the **US East (N. Virginia)** region.

74. ***Go to the left*** navigation pane, click **Transit Gateway Attachment**.

a. Select **LAB-VPC-Attachment**.

i. Select **Actions**.

ii. Select **Delete**.

a) When ***prompt for confirmation***, type **delete**.

iii. Select **Delete**.

b. Select **Shared-VPC-Attachment**.

i. Select **Actions**.

ii. Select **Delete**.

a) When ***prompt for confirmation***, type **delete**.

iii. Select **Delete**.

c. Select **Peering-with-WestUS02-Attachment**.

i. Select **Actions**.

ii. Select **Delete**.

iii. Select **Delete**.

**Note**: You can see the **State** of all the **Attachment** as **Deleting**.

75. ***Go to the left*** navigation pane, click **Transit Gateways**.

a. Select **US-East-01-TGW**.

i. Select **Actions**.

ii. Select **Delete**.

iii. Select **Delete**

**Note**: You can see the **State** of **Transit Gateway** as **Deleting**.

## Step 2: Terminate EC2 Instances from Region-1

76. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

77. Dropdown and select the **US East (N. Virginia)** region.

78. Click **Instances**.

a. Select the **WebApp Server-01** instances.

i. Click on **Instance state**.

ii. Select **Terminate instance**.

iii. Select **Terminate**.

### Step 3: Delete Security Groups from Region-1

79. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

80. Dropdown and select the **US East (N. Virginia)** region.

81. Select the **Security groups**.

   a. Select the **WebServer-SG**.

      i. Select **Actions**.

      ii. Select **Delete security groups**.

      iii. To confirm deletion, type **delete**.

      iv. Choose **Delete**.

### Step 4: Delete the Stack from Region-1

82. In the AWS Management Console, on the **Services** menu, click **CloudFormation**.

83. Dropdown and select the **US East (N. Virginia)** region.

   a. Select **Stack**.

      i. Select **LABVPC-01**.

         a) Select **Delete**.

         b) Select **Delete stack**.

      ii. Select **Shared-VPC**.

         c) Select **Delete**.

         d) Select **Delete stack**.

### Step 5: Delete Transit Gateway and Attachment from Region-2

84. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

85. Dropdown and select the **US West (Oregon)** region.

86. ***Go to the left*** navigation pane, click **Transit Gateway Attachment**.

a. Select **LAB-VPC02_R2-Attachment**.

i. Select **Actions**.

ii. Select **Delete**.

a) When ***prompt for confirmation***, type **delete**.

iii. Select **Delete**.

**Note**: You can see the **State** of all the **Attachment** as **Deleting**.

87. ***Go to the left*** navigation pane, click **Transit Gateways**.

a. Select **US-West-02-TGW**.

i. Select **Actions**.

ii. Select **Delete**.

a) When ***prompt for confirmation***, type **delete**.

iii. Select **Delete**

**Note**: You can see the **State** of **Transit Gateway** as **Deleting**.

**Step 4: Delete the Stack from Region-2**

88. In the AWS Management Console, on the **Services** menu, click **CloudFormation**.

89. Dropdown and select the **US West (Oregon)** region.

a. Select **Stack**.

b. Select **LAB-VPC02_R2**.

c. Select **Delete**.

b. Select **Delete stack**.